Dice Rolling Experiment - Printable Worksheet

Student worksheet without the interactive simulator

Year 11 Statistics

today

Name: ____ **Class:** _______________

Purpose

Use a fair six-sided die to investigate chance behaviour. Complete the tasks below after making observations from a class simulator, physical die, or other approved random experiment.

Part 1: Prediction

Before you start, answer the following questions.

1. Which face do you think will appear most often after many rolls?

2. Do you think the frequencies will be exactly equal? Explain why or why not.

Part 2: Record 20 Rolls

Record 20 rolls in the table below.

Roll number	Result	Roll number	Result	Roll number	Result	Roll number	Result
1	_________	6	_________	11	_________	16	_________
2	_________	7	_________	12	_________	17	_________
3	_________	8	_________	13	_________	18	_________
4	_________	9	_________	14	_________	19	_________
5	_________	10	_________	15	_________	20	_________

Part 3: Frequency Table

After 20 rolls, complete the summary table.

Face	Count	Experimental probability
1	_________	_________
2	_________	_________
3	_________	_________
4	_________	_________
5	_________	_________
6	_________	_________

Write the theoretical probability for each face of a fair die.

Part 4: Compare Results

1. Which face had the highest experimental frequency?

2. Which face had the lowest experimental frequency?

3. Were the results exactly the same as the theoretical probabilities? Explain why this happens.

Part 5: Reflection

How does the number of rolls affect how close the experimental probability gets to the theoretical probability of $\frac{1}{6}$?

Extension

Try 50 or 100 rolls and look for a pattern. Describe what happens to the bars or counts as the number of trials increases.